

Luigi Gargioni

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Lavoro: Brescia (Italia)

ESPERIENZA LAVORATIVA

Dottorato di RIcerca

Antares Vision [01/12/2022 – Attuale]

Città: Travagliato, Brescia | **Paese:** Italia

Telemedicina e Digitalizzazione dei Servizi Sanitari

Ingegnere sviluppo software - Libero professionista

Gargioni Luigi [12/2022 – Attuale]

Città: Brescia | **Paese:** Italia

• Sviluppo software Full-Stack
• Sito web: https://luigigargioni.github.io/

Scoutman

TieBreakTech [10/2019 – Attuale]

Città: Montichiari (Brescia) | **Paese:** Italia

Scoutman per la Lega Pallavolo Serie A Femminile presso il PalaGeorge di Montichiari (BS) e il PalaRadi di Cremona.

Attività svolte:
• Raccolta statistiche e collaborazione con gli addetti stampa.

Software Engineer - R&D Developer

Kline s.r.l. [16/11/2019 – 30/11/2022]

Città: Leno (Brescia) | **Paese:** Italia

Kline sviluppa e produce piattaforme software integrate, soluzioni tecnologiche e servizi per gli intermediari finanziari.

Dipartimento di Ricerca e Sviluppo - Attività svolte:
- Reengineering applicazione esistente con paradigma di programmazione funzionale e tecnologie web: JavaScript, TypeScript, Webpack e React (Redux, Styled Components, Jest, ...).
- Product Owner di due progetti in outsourcing con una società internazionale
- Selezione del personale

Precedentemente - Analista Programmatore:
- Analisi funzionale di procedure in ambito bancario e finanziario.
- Implementazione dei moduli in Delphi, Java e PL/SQL (Oracle DB).
- Attività di supporto ai clienti in relazione agli applicativi sviluppati e distribuiti.

Collaboratore

Punto Pizza [27/02/2011 – 29/12/2021]

Città: Gambara (Brescia) | **Paese:** Italia

Pizzeria d'asporto e al taglio.

Attività svolte:

- Aiuto pizzaiolo e addetto al bancone presso l'attività di famiglia

Sviluppatore di software

C&B Servizi Informatici [29/11/2016 – 28/06/2017]

Città: Casalmoro (Mantova) | **Paese:** Italia

Stage esterno all'università svolto durante il periodo di studi: finito il periodo stabilito, possibilità di assunzione non accettata per scelta di proseguire gli studi ed inizio del corso di laurea magistrale in ingegneria informatica. Occasionali collaborazioni a progetto nel 2020/2021.

Attività svolte:

- Attività di sviluppo e debug di software in ambiente Microsoft e Web: Visual Basic .NET, Visual Basic 6, ASP.NET, Javascript, Microsoft SQL Server e Access.
- Attività di installazione e manutenzione reti locali, server dipartimentali e apparati di rete.
- Affiancamento al personale in attività di installazione e manutenzione di personal computer e periferiche sia presso la nostra sede che presso la sede del cliente.

Addetto al bancone e cameriere

Camping Village San Francesco [30/05/2014 – 29/08/2014]

Città: Desenzano del Garda (Brescia) | **Paese:** Italia

Attività svolte:

- Addetto al bancone e cameriere per la stagione estiva

ISTRUZIONE E FORMAZIONE

Dottorato di Ricerca - Technology for Health

Università degli Studi di Brescia [01/12/2022 – Attuale]

Indirizzo: Via Branze 38, 25123 Brescia (Italia) | **Sito web:** https://tech4health.unibs.it/

Abilitazione alla professione di ingegnere dell'informazione

Università degli Studi di Brescia [22/07/2020 – 22/07/2020]

Indirizzo: Via Branze 38, 25123 Brescia (Italia) | **Sito web:** https://www.unibs.it/

Laurea Magistrale, Ingegneria Informatica

Università degli Studi di Brescia [2017 – 2019]

Indirizzo: Via Branze 38, 25123 Brescia (Italia) | **Sito web:** https://www.unibs.it/ | **Voto finale:** 107/110 | **Tesi:** A domain-independent environment for user-oriented programming of collaborative robots

Link: https://www.sciencedirect.com/science/article/abs/pii/S073658452100106X

Laurea Triennale, Ingegneria Informatica

Università degli Studi di Brescia [2012 – 2016]

Indirizzo: Via Branze 38, 25123 Brescia (Italia) | **Sito web:** https://www.unibs.it/ | **Tesi:** Verifica di vulnerabilità dei sistemi: concetti generali e analisi dello srumento OpenVAS

Diploma Liceo Scientifico Tecnologico

Istituto Istruzione Superiore Blaise Pascal [09/2007 – 06/2012]

Indirizzo: Via Solferino 92, 25025 Manerbio (Brescia) (Italia) | **Sito web:** https://www.iis-pascal.edu.it/ | **Tesi:** Concetti di chimica nucleare ed analisi dei disastri di Černobyl' e Fukushima

COMPETENZE LINGUISTICHE

Lingua madre: Italiano

Altre lingue:

Inglese

ASCOLTO B2 **LETTURA** B2 **SCRITTURA** B2

PRODUZIONE ORALE B2 **INTERAZIONE ORALE** B2

Francese

ASCOLTO A1 **LETTURA** A1 **SCRITTURA** A1

PRODUZIONE ORALE A1 **INTERAZIONE ORALE** A1

Livelli: A1 e A2: Livello elementare B1 e B2: Livello intermedio C1 e C2: Livello avanzato

COMPETENZE DIGITALI

Front-End React Redux JavaScript Bootstrap HTML5 CSS3 / Utilizzo base di git (git bash, gitlab, ...) / Node.Js / Appassionato di robotica, intelligenza artificiale e machine learning / Interfaccia uomo-macchina / Ricerca e Sviluppo

PUBBLICAZIONI

[2022]

A hybrid approach to user-oriented programming of collaborative robots The research reported in this paper proposes a new approach to collaborative robots that aims at improving the simplicity and efficiency of the programming task for non-technical users. It is grounded on three standpoints: (i) an elementary and disciplined paradigm for robot programming, called the simple programming journey, (ii) a hybrid interaction mode where robot tasks can be programmed using a natural language chat and, if necessary, can be completed and finalized through a block-based interface, and (iii) a robust cognitive match between the mental models of the user and the programming interface. The proposed approach has been implemented and tested through the development of a prototype programming environment called CAPIRCI, which can be tailored to different application domains through the definition of objects, locations, and actions. CAPIRCI has been tested by real users with a COBOTTA robot by DENSO WAVE Ltd. Two experimental tests have been carried out in order to validate the novel approach proposed and to assess its impact on end-user programming. The results obtained show that a hybrid approach exploiting both natural language dialogue and block-based interaction can help make the programming task easy and efficient for non-technical users.

Robotics and Computer-Integrated Manufacturing, Volume 73, February 2022

[2023]

Human-robot collaboration in healthcare: new programming and interaction techniques In the field of healthcare, collaborative robots can be utilized to enhance productivity and efficiency. This research will delve into the development and investigation of new techniques to enable effective interaction with collaborative robots. The focus will be on designing innovative techniques that can be easily understood and implemented by individuals with no technical knowledge in computer science or robotics. Collaborative robots can be used to automate repetitive tasks, allowing healthcare professionals to focus on more complex procedures. For instance, by leveraging this technology, it is possible to significantly increase the efficiency and speed of therapy preparation, ultimately improving patient outcomes. The study will explore various approaches to simplify and streamline the programming process, reducing the need for technical knowledge and expertise.

IS-EUD 2023, 9th International Symposium on End-User Development, 06-08 June 2023, Cagliari, Italy

[2023]

Can ChatGPT Support End-User Development of Robot Programs? In this paper, we investigate the use of OpenAI ChatGPT to improve the natural language understanding of an End-User Development environment, called CAPIRCI, supporting users neither expert in computer programming nor expert in robotics to create

programs for a collaborative robot. The integration of ChatGPT in CAPIRCI is studied to be transparent for the user, who will be allowed to check and correct that program description generated by ChatGPT, by interacting with an intuitive block-based interface, according to a Human-Centered Artificial Intelligence design approach.

CoPDA 2023, Cultures of Partecipation in the Digital Age, International workshop @ IS-EUD 2023

[2023]

Designing Human-Robot Collaboration for the Preparation of Personalized Medicines Advancements in robotics and automation technologies have the potential to enable breakthrough innovations in a variety of industries, and the pharmaceutical sector is no exception. The preparation of galenic formulations, involving the compounding and dispensing of medications, when personalized medicines are needed, e.g., to overcome allergy problems, is a critical process in the field of small scale pharmaceutical manufacturing. Traditionally, this process has relied solely on human expertise of pharmacists and their manual labor, which can be time-consuming, prone to errors, and subject to variations in quality. To overcome these limitations, the use of collaborative robots is envisaged in our project. A collaborative robot can in fact work with the pharmacist synergistically, by improving accuracy and increasing productivity. However, the main challenge is providing the pharmacists with an interactive system that supports them in robot programming. In this paper, we analyze the problem from the users' point of view and propose preliminary low-fidelity prototypes of an interactive system suitable to pharmacists' needs and skills.

GoodIT 2023, International Conference on Information Technology for Social Good, Lisbon, Portugal

[2023]

A Systematic Review on Pill and Medication Dispensers from a Human-Centered Perspective As medication adherence represents a critical challenge in healthcare, pill and medication dispensers have gained increasing attention as potential solutions to promote adherence and improve patient outcomes. Following the PRISMA (Preferred Reporting Items for Systematic reviews and Meta-Analyses) methodology, we carried out a systematic literature review on papers indexed in Scopus, which present solutions for pill or medication dispensers. Given the importance of user acceptance for these solutions, the research questions of the survey are driven by a human-centered perspective. We first provide an overview of the different solutions, classifying them according to their stage of development. We then analyze each solution considering its hardware/software architecture. Finally, we review the characteristics of user interfaces designed for interacting with pill and medication dispensers and analyze the involvement of different types of users in dispenser management. On the basis of this analysis, we draw findings and indications for future research that are aimed to provide insights to healthcare professionals, researchers, and designers who are interested in developing and using pill and medication dispensers.

Journal of Healthcare Informatics Research, Springer - Peer Review Stage

RETI E AFFILIAZIONI

[11/2022 – Attuale] Brescia

Associazione Alumni Università degli Studi di Brescia

Link: https://alumni.unibs.it/documents/badges/badge.php?hash=02608da02a1dbd8ef89a8bd5b40403df2a766f75

PROGETTI

[12/2020 – 08/2022]

Gestionale per carpenteria metallica Gestionale per carpenteria metallica sviluppato con tecnologie web:
• Front-end: Typescript, Electron, React, Redux, Webpack, Ant Design, Styled-Components
• Back-end: Typescript, Node.js, Express, Sequelize, Morgan, JWT, bcrypt
• Database: MariaDB

• Controllo di versione con Git (GitHub)
• Supporto (HW/SW): Raspberry Pi, GitHub Releases, Husky, PM2

[02/2019 – 10/2019]
CAPIRCI (Chat And Program Industrial Robots through Convenient Interaction) CAPIRCI (Chat And Program Industrial Robots through Convenient Interaction), un'applicazione web multimodale che supporta gli utenti finali, senza alcuna competenza in informatica, per definire e modificare le attività da eseguire da robot collaborativi. L'applicazione fornisce due modalità di interazione, la prima basata su un'interfaccia di chat, la seconda che presenta un linguaggio di programmazione visuale ispirato a soluzioni basate su blocchi, ma adattato al dominio specifico.
Tecnologie utilizzate: Python, Django, OpenCV, Javascript, Ajax, Css, Standford Parser, ORiN e B-cap.
Robot utilizzato: Denso Cobotta con camera Canon N10-W02 e Denso VS.

[08/2018 – 05/2019]
PallavoloRemedello.it Sito Web per la squadra di pallavolo A.S.D. Pallavolo Remedello.
Tecnologie utilizzate: Javascript, jQuery, Ajax, PHP e CSS.

ONORIFICENZE E RICONOSCIMENTI

[02/2023] MongoDB
MongoDB Node.js Developer Path
Link: https://learn.mongodb.com/learn/learning-path/mongodb-nodejs-developer-path

[01/2023] Postman
Postman API Fundamentals Student Expert ID credenziale: WIoUhlzfQOqVq57BMYKK_Q
Link: https://badgr.com/public/assertions/WIoUhlzfQOqVq57BMYKK_Q?identity__email=l.gargioni@studenti.unibs.it

[08/2022] MIT Open Learning
Machine Learning
Link: https://openlearninglibrary.mit.edu/courses/course-v1:MITx+6.036+1T2019/course/

[04/2022] École Polytechnique
How to Write and Publish a Scientific Paper (Project-Centered Course) ID credenziale: 46CKHJA5BZRE
Link: https://www.coursera.org/account/accomplishments/verify/46CKHJA5BZRE

[04/2022] Fastweb Digital Academy
Intelligenza Artificiale e sostenibilità ID credenziale: 63382067693d40f6c9f071c284da76a972873f62
Link: https://learn.fastwebdigital.academy/badges/badge.php?hash=63382067693d40f6c9f071c284da76a972873f62

[11/2021] University of Helsinki
Elements of AI ID credenziale: 7kas3px5o4q
Link: https://certificates.mooc.fi/validate/7kas3px5o4q

[09/2020] Stanford University
Machine Learning ID credenziale: G5J6SZ2UXRDX

Link: https://www.coursera.org/account/accomplishments/verify/G5J6SZ2UXRDX

[09/2020] 3CX

Basic Certification ID credenziale: ezftPKvV8p

Link: https://customer.3cx.com/documents/CertificationSharing.aspx?c=ezftPKvV8p

[04/2020] Cisco Systems

Cisco Networking Academy

- Intro to IoE
- Introduction to Cybersecurity
- Cybersecurity Essentials
- NDG Linux Essentials
- NDG Linux Unhatched

Link: https://www.youracclaim.com/badges/a4cdaf40-7ae7-4422-a89b-5b9ca09f7836

[03/2012] Trinity College London

ISE I Integrated Skills in English - B1 of the CEFR

[06/2011] AICA - Associazione Italiana per l'Informatica e il Calcolo Automatico

European Computer Driving License (ECDL)

HOBBY E INTERESSI

Pallavolo

Calcio

Softair

Lettura